Exam 2

ECE 2201: Circuit Analysis I

Summer 2025

July 3, 2025

1. This exam is closed book, closed notes and a crib sheet has been provided to you with the exam. Print your name below.
2. Show all work on these pages. Show all work necessary to complete the problem and explain the steps in your solution. A solution without the appropriate work shown will receive no credit. A solution which is not given in a reasonable order will lose credit.
3. Show all units in solutions, intermediate results, and figures. Units in the exam will be included between square brackets.
4. If the grader has difficulty following your work because it is messy or disorganized, you will lose credit.
5. Do not use red ink. Do not use red pencil.
6. You will have 75 minutes to work on this exam.

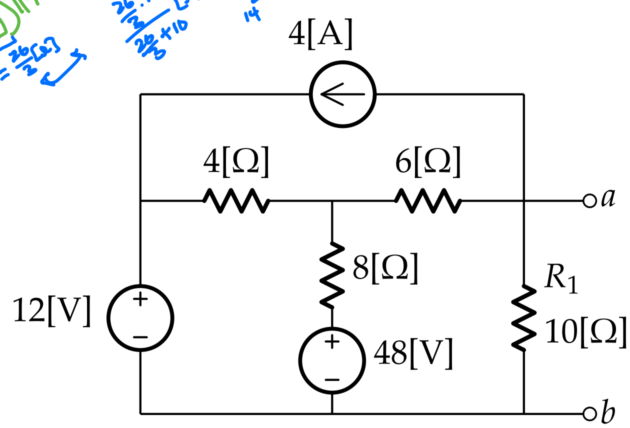
Name: Solution ID: __________

1. ________ /25
2. ________ /25
3. ________ /25
4. ________ /25

Problem 1

Note: to receive credit, you must use superposition to solve this problem.

a. Use superposition to solve for V_{ab} and write each solution for V_{ab}', V_{ab}'', V_{ab}''' and V_{ab} in the spaces below
b. How much power is dissipated in R_1?



alternative by:

$\frac{4\cdot 8}{4+8}[\Omega]=\frac{8}{3}[\Omega]$

$V_{ab} = -4[A]\left(4[\Omega]\|8[\Omega]+6[\Omega]\right)\|10[\Omega]$

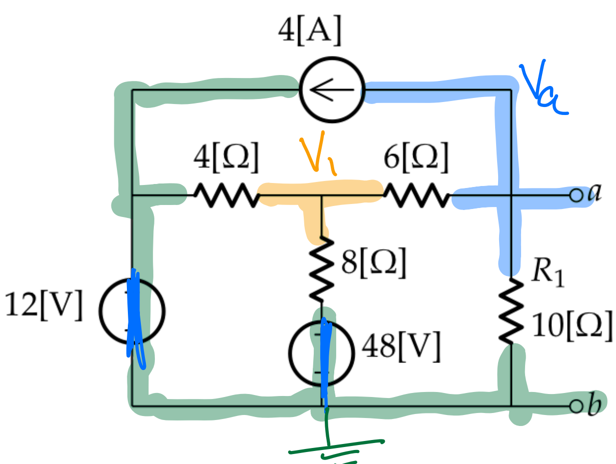
$\left(\frac{8}{3}+\frac{18}{3}\right)[\Omega]=\frac{26}{3}[\Omega]$

$\frac{\frac{26}{3}\cdot 10}{\frac{26}{3}+10}[\Omega] = \frac{65}{14}[\Omega]$

$= -4[A]\cdot\frac{65}{14}[\Omega]$

$= -\frac{130}{7}[V]$

① Current Source



KCL

$$\left[\frac{V_1}{4[\Omega]}+\frac{V_1}{8[\Omega]}+\frac{V_1-V_a}{6[\Omega]}=0\right]\times 24[\Omega]$$

$$6V_1+3V_1+4V_1-4V_a=0$$

$$13V_1-4V_a=0$$

KCL

$$\left[4[A]+\frac{V_a-V_1}{6[\Omega]}+\frac{V_a}{10[\Omega]}=0\right]\times 30[\Omega]$$

$$120[V]+5V_a-5V_1+3V_a=0$$

$$-5V_1+8V_a=-120[V]$$

$$26V_1-8V_a=0$$
$$-5V_1+8V_a=-120[V]$$

$$21V_1+0=-120[V]$$

$$V_1=\frac{-120}{21}[V]=-\frac{40}{7}[V]$$

$$V_a=\frac{13}{4}\cdot V_1=\frac{13}{\cancel{4}}\left(-\frac{\cancel{40}^{10}}{7}[V]\right)$$

$$=-\frac{130}{7}[V] \qquad V_{ab}'=-\frac{130}{7}[V]$$

$V_{ab}' =$ $-\frac{130}{7}[V]$ $=-18.6[V]$

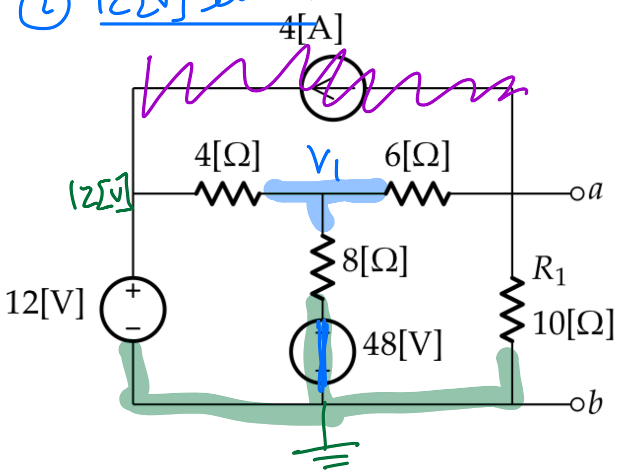
$V_{ab}'' =$ $\frac{30}{7}[V]$ $=4.3[V]$

$V_{ab}''' =$ $\frac{60}{7}[V]$ $=8.6[V]$

$V_{ab} =$ $-\frac{40}{7}[V]$ $=-5.7[V]$

$p_{R1} =$ $3.3[W]$

② 12[V] source



KCL

$$\left[\frac{V_1 - 12[V]}{4[\Omega]} + \frac{V_1}{8[\Omega]} + \frac{V_1}{16[\Omega]} = 0\right] \times 16[\Omega]$$

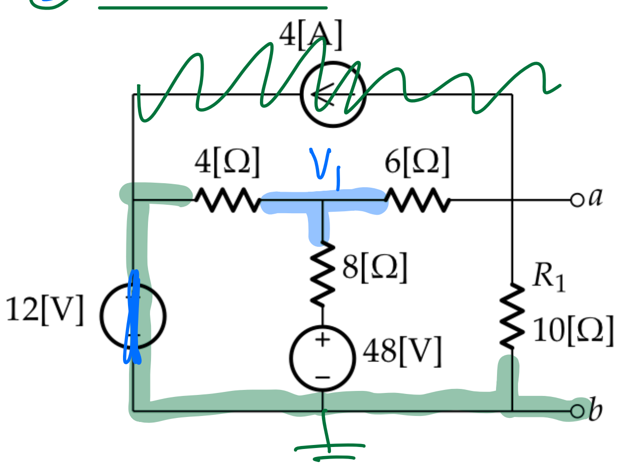
$$4V_1 - 48[V] + 2V_1 + V_1 = 0$$

$$7V_1 = 48[V] \Rightarrow V_1 = \frac{48}{7}[V]$$

Voltage Div.:

$$V_{ab}'' = V_1 \cdot \frac{10[\Omega]}{6[\Omega] + 10[\Omega]} = \frac{\cancel{48}^{3}}{7}[V] \cdot \frac{10}{\cancel{16}} = \frac{30}{7}[V]$$

③ 48[V] source



KCL

$$\left[\frac{V_1}{4[\Omega]} + \frac{V_1 - 48[V]}{8[\Omega]} + \frac{V_1}{16[\Omega]} = 0\right] \times 16[\Omega]$$

$$4V_1 + 2V_1 - 96[V] + V_1 = 0$$

$$7V_1 = 96[V] \Rightarrow V_1 = \frac{96}{7}[V]$$

Voltage Div.:

$$V_{ab}''' = V_1 \cdot \frac{10[\Omega]}{6[\Omega] + 10[\Omega]} = \frac{\cancel{96}^{6}}{7}[V] \cdot \frac{10}{\cancel{16}}$$

$$V_{ab}''' = \frac{60}{7}[V]$$

$$V_{ab} = V_{ab}' + V_{ab}'' + V_{ab}'''$$

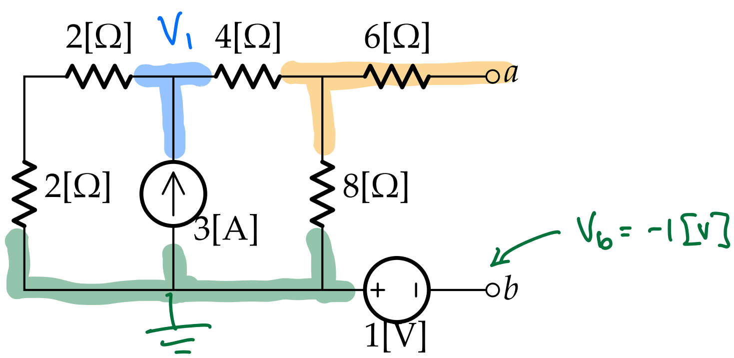
$$= \left(-\frac{130}{7} + \frac{30}{7} + \frac{60}{7}\right)[V]$$

$$= -\frac{40}{7}[V]$$

$$P_{R_1} = \frac{(V_{ab})^2}{R_1} = \frac{\left(-\frac{40}{7}\right)^2}{10}[W] = \frac{1600}{49 \cdot 10}[W] = \frac{160}{49}[W] = 3.3[W]$$

Problem 2

a. Find the Thévenin equivalent circuit at terminals (a,b) in the circuit below.
b. If a load resistor is placed between terminals (a,b), what resistance value would ensure maximum power transfer?



KCL:

$$\left[\frac{V_1}{4[\Omega]} - 3[A] + \frac{V_1}{12[\Omega]} = 0\right] \times 12[\Omega]$$

$$3V_1 - 36[V] + V_1 = 0 \Rightarrow 4V_1 = 36[V] \Rightarrow V_1 = 9[V]$$

$$V_a = V_1 \cdot \frac{8[\Omega]}{4[\Omega]+8[\Omega]} = \cancel{9}^{3}[V] \cdot \frac{\cancel{8}^{2}}{\cancel{12}_{4}} = 6[V]$$

$$V_{Th} = V_{ab} = V_a - V_b = 6[V] - (-1[V]) = \underline{\underline{7[V]}}$$

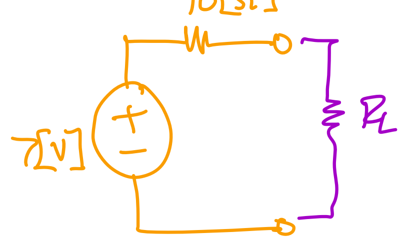
$R_{eq} \rightarrow$ turn off all indep. sources:

$$R_{eq} = (2[\Omega]+2[\Omega]+4[\Omega]) \,||\, 8[\Omega] + 6[\Omega]$$

$$= 8[\Omega] \,||\, 8[\Omega] + 6[\Omega] = 4[\Omega] + 6[\Omega]$$

$$\underline{R_{eq} = 10[\Omega]}$$

Thevenin Equiv.:

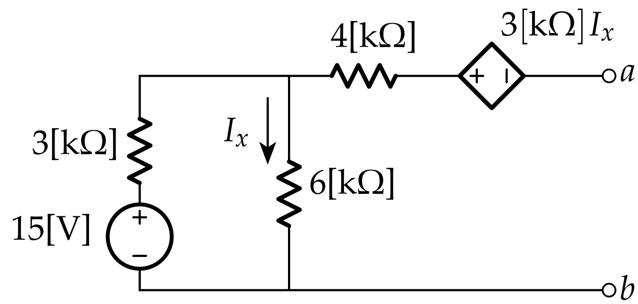


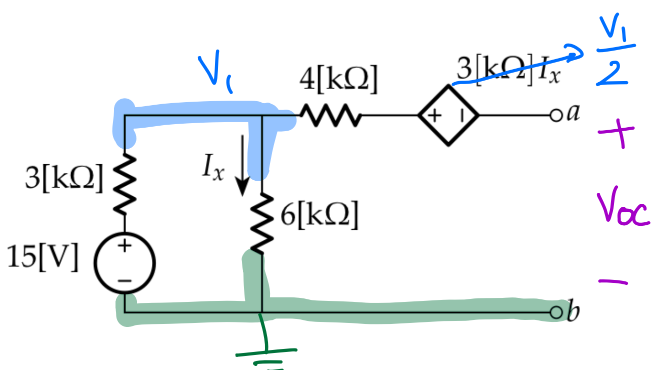
For max. power transfer,

$R_L = R_{Th} = 10[\Omega]$

Problem 3

Find the Thévenin equivalent circuit at terminals (a,b) for the circuit below





Note that $I_x = \frac{V_1}{6[k\Omega]}$

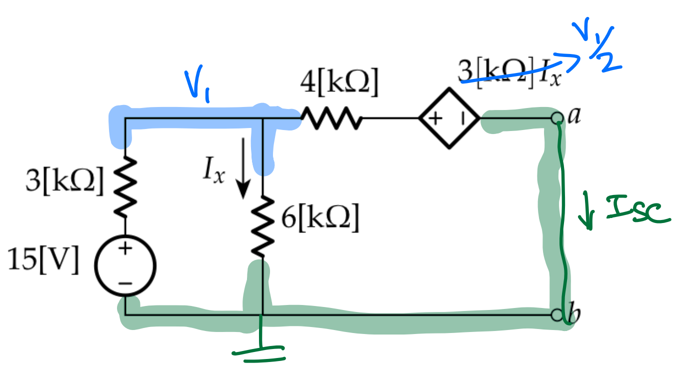
and so $3[k\Omega]\cdot I_x = \frac{3}{6}V_1 = \frac{V_1}{2}$

KCL: $\frac{V_1 - 15[V]}{3[k\Omega]} + \frac{V_1}{6[k\Omega]} = 0 \Rightarrow 3V_1 = 30[V]$

$V_1 = 10[V]$

Alternatively, we can use a Voltage Divider: $V_1 = 15[V]\cdot\frac{6[k\Omega]}{3[k\Omega]+6[k\Omega]} = \cancel{15}\,5[V]\cdot\frac{\cancel{6}\,2}{\cancel{9}\,\cancel{3}} = 10[V]$

$V_{oc} = V_1 - \frac{V_1}{2} = \frac{V_1}{2} = 5[V] = V_{Th}$



KCL:

$$\left[\frac{V_1 - 15[V]}{3[k\Omega]} + \frac{V_1}{6[k\Omega]} + \frac{V_1 - \frac{V_1}{2}}{4[k\Omega]} = 0\right]\times 24[\Omega]$$

$8V_1 - 120[V] + 4V_1 + 3V_1 = 0$

$15V_1 = 120[V]$

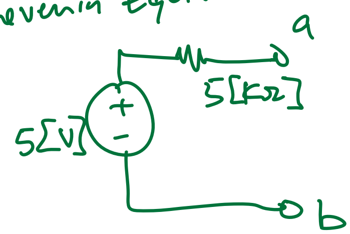
$V_1 = 8[V]$

$$I_{sc} = \frac{V_1 - \frac{V_1}{2}}{4[k\Omega]} = \frac{\frac{V_1}{2}}{4[k\Omega]} = \frac{\frac{8[V]}{2}}{4[k\Omega]}$$

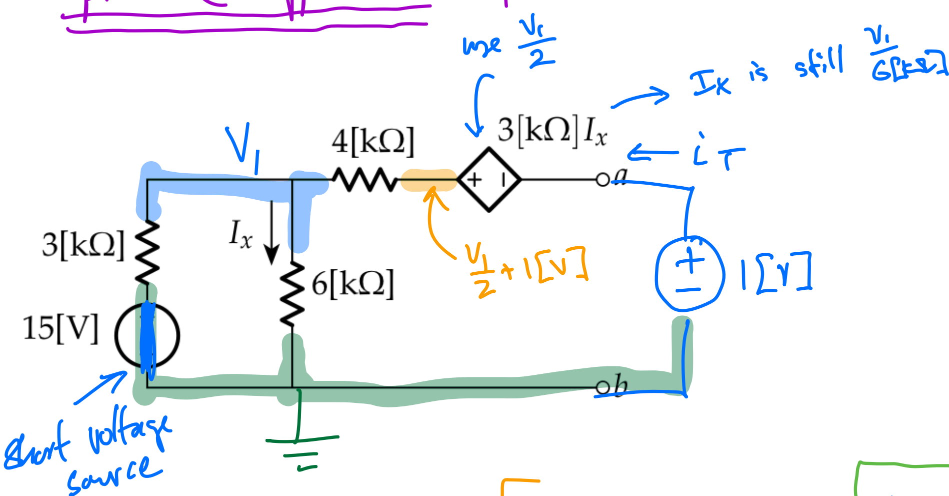
$I_{sc} = 1[mA]$

$$R_{Th} = \frac{V_{oc}}{I_{sc}} = \frac{5[V]}{1[mA]} = 5[k\Omega]$$

Thevenin Equiv.:



Optional Approach: find R_{Th} using a test voltage source



$$\left[\frac{V_1}{3[k\Omega]} + \frac{V_1}{6[k\Omega]} + \frac{\overbrace{V_1 - \left(\frac{V_1}{2} + 1[V]\right)}^{\frac{V_1}{2} - 1[V]}}{4[k\Omega]} = 0\right] \times 12[\Omega]$$

$$\left[4V_1 + 2V_1 + \frac{3}{2}V_1 - 3[V] = 0\right] \times 2$$

$$8V_1 + 4V_1 + 3V_1 - 6[V] = 0$$

$$15V_1 = 6[V]$$

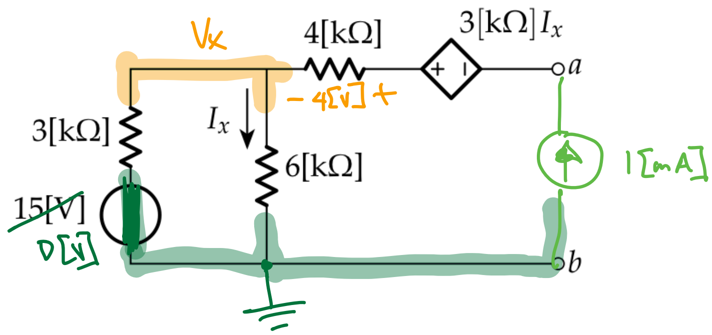
$$V_1 = \frac{2}{5}[V]$$

$$i_T = \frac{1[V] + \frac{V_1}{2} - V_1}{4[k\Omega]} = \frac{1[V] + \frac{2}{10}[V] - \frac{2}{5}[V]}{4[k\Omega]}$$

$$= \frac{\frac{\cancel{8}\,2}{10}[V]}{4[k\Omega]} = \frac{1}{5}\,[mA]$$

$$R_{Th} = \frac{V_T}{i_T} = \frac{1[V]}{\frac{1}{5}[mA]} = \underline{\underline{5[k\Omega]}}$$

Another optional Approach: use a test current source



$$\frac{V_x}{3[k\Omega]} + \frac{V_x}{6[k\Omega]} - 1[mA] = 0$$

$$2V_x + V_x = 6[V]$$

$$V_x = 2[V]$$

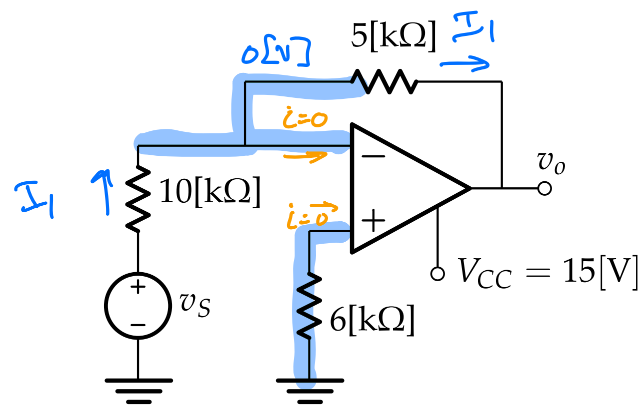
$$V_{ab} = V_x + 4[V] - \frac{V_x}{6[k\Omega]} \cdot 3[k\Omega]$$

$$= 2[V] + 4[V] - 1[V] = 5[V]$$

$$R_{Th} = \frac{V_{ab}}{1[mA]} = \frac{5[V]}{1[mA]} = 5[k\Omega]$$

Problem 4

a. Determine the output voltage (v_o) in the circuit below in terms of v_s
b. Specify the linear range for v_s



(a) $I_1 = \frac{V_s}{10[k\Omega]}$

$0[V] - V_o = I_1 5[k\Omega]$

$V_o = -\frac{V_s}{10[k\Omega]} \cdot 5[k\Omega] = -\frac{1}{2}V_s$

$$\boxed{V_o = -\frac{1}{2}V_s}$$

(b) $-15[V] < V_o < 15[V]$

$-15[V] < -\frac{1}{2}V_s < 15[V]$

$$\boxed{-30[V] < V_s < 30[V]}$$